

Lecture 4

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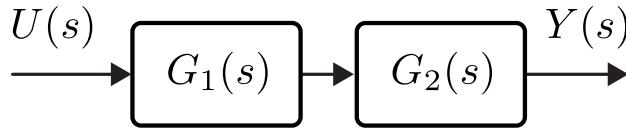
This lecture note is originally prepared by Dr. Mert Ankarali for METU EE-302.

(Source: <https://github.com/mertankarali/Lecture-Notes>)

4.1 Block Diagrams & Simplifications

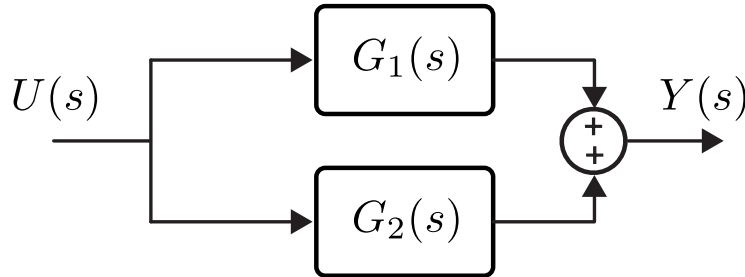
4.1.1 Fundamental Block Diagram Topologies

Cascaded (Series) Block Diagrams



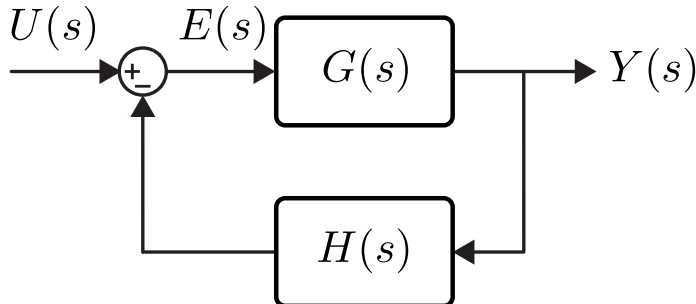
$$\frac{Y(s)}{U(s)} = \bar{G}(s) = G_1(s)G_2(s)$$

Parallel Block Diagrams



$$\frac{Y(s)}{U(s)} = \bar{G}(s) = G_1(s) + G_2(s)$$

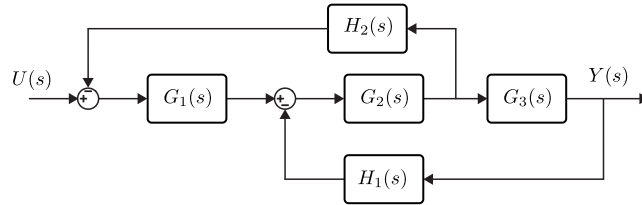
Negative Feedback Loop



$$\begin{aligned} E(s) &= U(s) - H(s)Y(s) \\ E(s)(1 + H(s)G(s)) &= U(s) \\ \frac{Y(s)}{U(s)} &= \bar{G}(s) = \frac{G(s)}{1 + H(s)G(s)} \end{aligned}$$

4.1.2 Examples

Ex 1: Simplify the following block-diagram topology



Solution:

